I. Introduction to Electronic Payment Systems (EPS)

Electronic Payment System:

A system that allows money to be paid or transferred electronically – without cash or cheques – between buyer and seller, using computers, networks, banks and payment instruments.

Meaning

- Payment is made by electronic message, card, app or digital token.
- Needed in e-commerce because buyer and seller are not face to face.
- Involves customer, merchant, issuing bank, acquiring bank, payment gateway and a clearing network.

Features of a Good EPS

- **Security:** protects money and data from fraud.
- **Authentication:** verifies identity of payer (PIN, OTP, biometrics).
- **Integrity & confidentiality:** amount and details cannot be altered or read by others.
- **Convenience & speed:** quick, easy, anytime.
- **Low transaction cost:** suitable for small payments (micro-payments).
- **Acceptability:** widely accepted by merchants and banks.
- **Anonymity / privacy:** in some systems (cash-like).
- **Reliability & interoperability:** works across banks and apps.

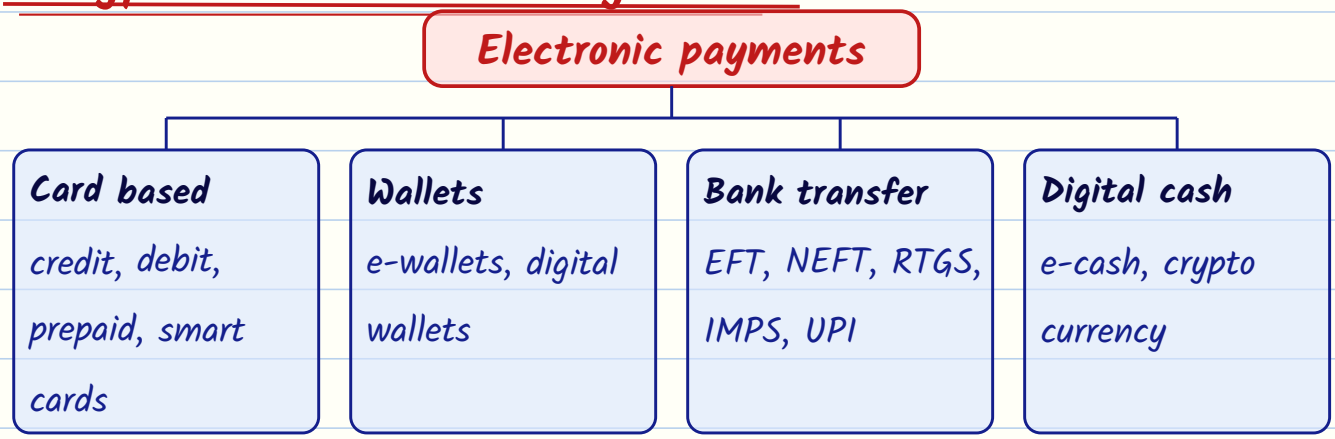
Advantages of EPS

- Faster payment and settlement; reduced cash handling.
- Transparent record of every transaction.
- Supports online shopping and global trade.
- Less risk of theft, fake notes and counting errors.
- Supports Digital India and financial inclusion.

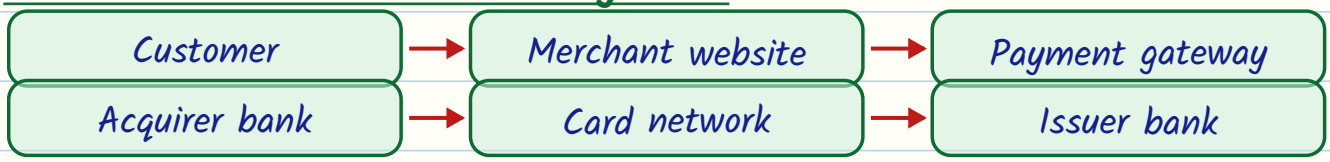
Disadvantages

- Cyber fraud, phishing, card cloning.
- Needs internet and a smartphone; technical failure.
- Transaction charges in some modes.
- Privacy - spending is tracked.

2. Types of Electronic Payments



Parties in an Online Card Payment



- **Payment gateway:** secure link that carries card data (Razorpay, PayU, CCAvenue, Paytm).
- **Acquirer bank:** merchant's bank.
- **Issuer bank:** customer's bank that gave the card.
- **Card network:** Visa, Mastercard, RuPay - route and settle the payment.

3. Credit Cards

Credit Card:

A plastic / virtual card issued by a bank that lets the holder buy now and pay later, within a sanctioned credit limit; the bank lends the money.

How It Works

- Bank sets a credit limit and billing cycle (about 30 days).
- Customer swipes / enters card details; merchant is paid by the bank.
- Customer receives a monthly statement and pays the bill by the due date.
- If only the minimum amount is paid or paid late, heavy interest (about 36 - 42 % p.a.) is charged.

Features

- Card number (16 digits), expiry date, CVV, name; chip and PIN or OTP.
- Interest-free credit period (up to about 50 days).
- Reward points, cashback, EMI options, travel / lounge benefits.
- International acceptance.

Advantages

- Buy first, pay later; emergency credit.
- Rewards, discounts and EMI facility.
- Builds credit history (CIBIL score).
- Purchase protection and dispute facility.

Disadvantages

- Overspending and debt trap; high interest and late fees.
- Annual fee and hidden charges.
- Fraud risk if details leak.

4. Debit Cards

Debit Card:

A card linked to the holder's savings / current account; the purchase amount is deducted directly from the account balance at the time of payment.

- Issued by banks: Visa, Mastercard, RuPay (e.g. SBI, HDFC, PNB cards).
- Used at ATM, POS terminals and online with PIN / OTP.

Features

- Instant debit; no borrowing.
- Daily withdrawal and spending limits.
- Works with 2-factor authentication.
- Contactless (tap and pay, NFC) in many cards.

Advantages

- No debt or interest; spending within balance.
- Easy cash withdrawal and cashless shopping.
- Usually no joining fee; simple to get.

Disadvantages

- No credit period; money leaves the account at once.
- Fraud directly affects bank balance.
- Fewer rewards and protection than credit cards.
- Payment fails if balance is low.

Credit Card vs Debit Card

Basis	Credit Card	Debit Card
Money used	Bank's money (loan)	Own money in account
Payment	Later - bill each month	Immediate deduction
Limit	Credit limit by bank	Account balance / daily limit

Interest	Charged if unpaid	None
Eligibility	Income and credit score check	Any account holder
Debt risk	High	Nil
Rewards	More rewards and benefits	Few offers
Fraud loss	Bank money first, dispute possible	Own money is lost first
Credit score	Improves with timely payment	No effect

★ Prepaid cards / gift cards: money is loaded first and used later - no credit, no bank account needed.

5. Smart Cards

Smart Card:
A plastic card with an embedded microchip (microprocessor + memory) that can store and process data securely for identity, payment and access.

Types

- **Contact smart card:** chip touches the reader (EMV chip cards, SIM).
- **Contactless card:** works by radio waves (RFID / NFC) - metro cards, tap-and-pay.
- **Hybrid / dual-interface:** both contact and contactless.

Features

- Stores money, personal data, keys and certificates in the chip.
- Strong security - PIN, encryption, tamper-proof chip.
- Offline transaction possible.
- Reusable and rechargeable (stored-value cards).

Uses

- Banking: EMV debit / credit cards.
- Transport: metro, bus, toll (FASTag uses RFID).
- Identity: Aadhaar, e-passport, employee ID.
- Telecom: SIM; health card; campus card; loyalty cards.

Advantages

- Safer than magnetic stripe; hard to clone.
- Quick tap payments; reduces cash.
- Many applications on one card.

Disadvantages

- Card reader needed; cost of issue.
- Lost card = loss of stored balance.
- Compatibility problems between systems.

6. E-Wallets and Digital Wallets

E-Wallet (Digital Wallet):
A software / app-based prepaid account that stores money or payment details electronically so that the user can pay, transfer and receive money using a phone or computer.

- Examples: Paytm, PhonePe wallet, Amazon Pay, MobiKwik, Freecharge, Google Pay, Apple Pay, Samsung Pay.

Types

- **Closed wallet:** only for the issuer's own products (Amazon Pay balance).
- **Semi-closed wallet:** many merchants, but no cash withdrawal (Paytm, Mobikwik).
- **Open wallet:** issued by banks; allows cash withdrawal and bank

transfer.

Features and Usage

- Load money by card, net banking or UPI.
- Pay via QR code, mobile number, NFC.
- Bill payment, recharge, ticket booking.
- Cashback, coupons, wallet-to-wallet transfer.
- KYC (Know Your Customer) needed for higher limits - regulated by RBI.

Advantages

- Fast, no need to enter card details each time.
- Offers and cashback; easy splitting of bills.
- Works for small value payments.

Disadvantages

- Limit on balance and transfers.
- Needs smartphone and internet; risk if phone is lost.
- Money may get stuck if the wallet company fails.

7. Smart Cards vs Digital Wallets

Basis	Smart Card	Digital Wallet
Form	Physical chip card	Software / mobile app
Storage	Chip in the card	Cloud / phone
Internet	Not always required	Usually required
Security	PIN + chip encryption	PIN, OTP, biometrics
Use	Offline and online	Mostly online / QR
Loss risk	Physical card loss	Phone loss
Examples	EMV card, metro card	Paytm, PhonePe, Google Pay

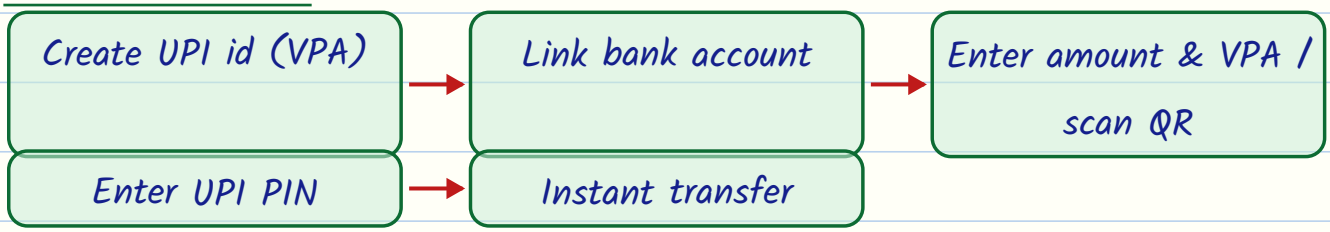
8. UPI - Unified Payments Interface

UPI:

A real-time payment system developed by NPCI (National Payments Corporation of India) that instantly transfers money between bank accounts through a mobile app using a Virtual Payment Address (VPA).

- Launched in 2016 by NPCI under RBI guidance; works 24 x 7 x 365.
- Apps: PhonePe, Google Pay, Paytm, BHIM, Amazon Pay and bank apps.

How UPI Works



Features

- Single app for many bank accounts; one mobile number = identity.
- Pay by VPA (name@bank), QR code, phone number.
- Send and request money (collect request).
- Two-factor security: device binding + UPI PIN.
- Free for most peer-to-peer payments.
- UPI Lite, UPI 123PAY (feature phones), international UPI.

Advantages

- Instant, simple, 24 x 7; no need to share account number or card.
- Interoperable between all banks; low cost.
- Supports merchants, bill pay, mandates (auto-pay).

Disadvantages

- Needs internet; server downtime sometimes.
- Fraud through fake collect requests and QR scams.
- Per-transaction limits (usually Rs. 1 lakh).

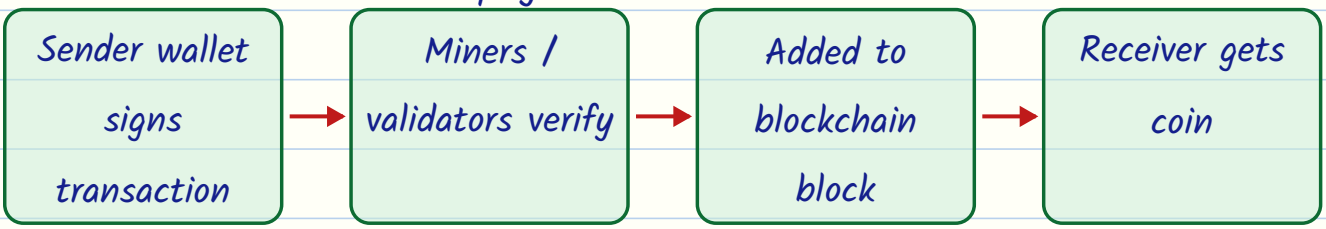
9. Crypto Currencies

Cryptocurrency:
A digital / virtual currency secured by cryptography, created and recorded on a decentralised blockchain, with no central bank or government control.

- Examples: Bitcoin (2009), Ethereum, Ripple (XRP), Litecoin, Tether (stable coin).

Features

- **Decentralised:** no single authority; peer-to-peer network.
- **Blockchain:** public ledger of all transactions, in linked blocks.
- **Cryptographic security:** public / private keys, hashing.
- **Limited supply:** e.g. Bitcoin max 21 million.
- **Pseudonymous & global:** transfers anywhere in minutes.
- **Irreversible:** confirmed payment cannot be reversed.



Advantages

- Low-cost cross-border payments; no middlemen.
- Strong security and transparency.
- Open to the unbanked.

Disadvantages

- Very volatile price; speculation risk.
- Used for illegal activity; scams and exchange hacks.
- Lost private key = lost coins.
- Regulation is uncertain; in India it is not legal tender - gains taxed at 30 % + 1 % TDS; RBI is developing the digital rupee (e-Rupee, a CBDC).

10. Electronic Fund Transfer (EFT)

EFT:

The electronic transfer of money from one bank account to another - within one bank or between banks - using computer networks, without paper cheques or cash.

Role of EFT in Business Transactions

- Salary and wage payment (payroll), pension, dividends.
- Supplier / vendor payments and customer receipts.
- Tax payment (GST, income tax), utility and loan EMIs.
- E-commerce settlement between buyer, seller and payment gateway.
- Reduces cash handling, float time, paperwork and errors.

Modes of EFT in India

Mode	Key points
NEFT	Batch based, 24 x 7; no minimum limit; small to medium amounts
RTGS	Real-time gross settlement; minimum Rs. 2 lakh; large value
IMPS	Instant 24 x 7 via mobile / net banking
UPI	Instant, via VPA / QR; PIN based

ECS / NACH	Bulk recurring credit / debit - salary, EMI, SIP
ATM / POS	Card based cash withdrawal and purchase

Advantages

- Fast and safe; lower cost; easy record keeping and audit.
- Reduces cheque bouncing and delays.

Disadvantages

- Wrong account number can cause wrong credit; reversal is difficult.
- Needs bank account and internet; cyber risks.

Smart Card vs EFT (Learning Outcome)

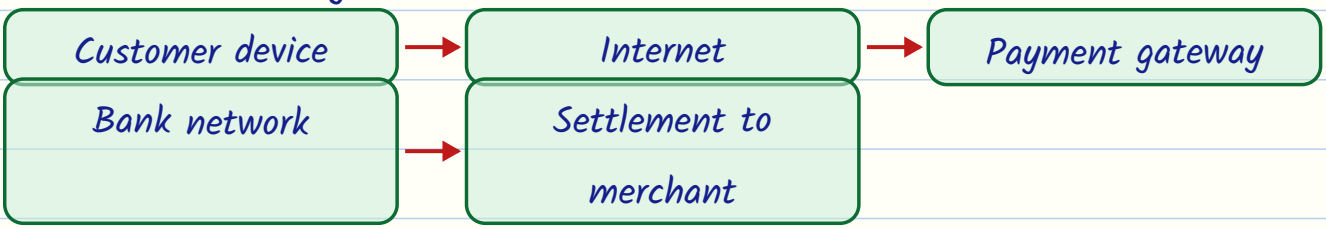
Basis	Smart Card	EFT
Nature	A payment instrument / device	A method of moving funds between accounts
Medium	Chip card with stored data	Bank network (NEFT, RTGS, IMPS)
Use	Point-of-sale, transport, ID	Salary, vendor, large business payments
Value	Small to medium	Small to very large
Money source	Stored value or linked account	Directly from bank account
Offline use	Possible	Needs network

II. Infrastructure Requirements for E-Payments

- **Hardware:** servers, POS machines, ATMs, smartphones, card readers, NFC / QR scanners.
- **Software:** payment gateway, banking software (CBS), merchant

apps, shopping-cart software.

- **Networks:** reliable internet, VPN / leased lines, SWIFT, NPCI network (UPI, IMPS, RuPay).
- **Security tools:** SSL / TLS, encryption, firewalls, tokenisation, OTP / biometrics, digital certificates.
- **Banking set-up:** merchant account, acquiring bank, issuer banks, clearing house.
- **Standards:** EMV chip standard, PCI-DSS (card data security), ISO 8583 messages.
- **Support:** call centre, dispute / chargeback system, backup and disaster recovery.



12. Regulatory Aspects of E-Payments

Key Authorities and Laws in India

- **RBI (Reserve Bank of India):** regulates banks, wallets (PPI guidelines), card rules, two-factor authentication, tokenisation of cards.
- **NPCI:** operates UPI, IMPS, RuPay, NACH, FASTag.
- **Payment and Settlement Systems Act, 2007:** legal base for payment systems; RBI authorises operators.
- **Information Technology Act, 2000:** recognises electronic records and digital signatures; cyber-crime offences.
- **Consumer Protection Act / E-Commerce Rules 2020:** protects

online buyers.

- **PMLA & KYC norms:** anti-money laundering checks.
- **Data protection:** Digital Personal Data Protection Act, 2023; RBI data storage rules.
- **PCI-DSS:** international security standard for card data.

Consumer Protection Measures

- Report unauthorised transaction quickly: zero liability if reported within 3 working days (as per RBI rules).
- Two-factor authentication, OTP, card tokenisation.
- Grievance redressal through bank, RBI Ombudsman and NPCI.

★ Tip: Never share OTP, PIN, CVV or UPI PIN with anyone.

13. Quick Revision - Unit III

- EPS = payment without cash. Types: cards, wallets, UPI, EFT, crypto, smart cards.
- Credit card = pay later (bank loan); debit card = pay now from own account.
- Smart card = chip card; EFT = account-to-account bank transfer (NEFT, RTGS, IMPS).
- E-wallet = prepaid app (closed, semi-closed, open). UPI = instant bank-to-bank via NPCI.
- Crypto = blockchain based decentralised digital money; risky and not legal tender in India.
- Infrastructure: servers, gateway, network, security, banks. Regulators: RBI, NPCI, IT Act, PSS Act.

Likely Exam Questions

- Explain types of electronic payment systems.
- Distinguish between credit card and debit card.
- Compare smart cards and EFT.
- Explain UPI and digital wallets with features.
- Discuss infrastructure and regulatory aspects of e-payments.